

Purple Tech Challenge 2026 - Problem Statement Summary

1. The Core Objective

You need to build an end-to-end Store Intelligence system for a retail chain. Like online stores track clicks, your job is to track offline customer behavior using raw CCTV footage.

2. The Provided Dataset

- CCTV Clips: 20-minute video clips from 5 stores, each with 3 camera angles (Entry/Exit, Main Floor, Billing Area). 1080p, 15fps, anonymized faces/branding.
- Store Layout (store_layout.json): Defines the zones, camera coverage, and open hours.
- POS Transactions (pos_transactions.csv): Timestamped purchase records. Must correlate purchases to customers by time and zone.
- Sample Events & Assertions: Examples of expected output format and tests your API must pass.

3. The 4-Stage Pipeline You Must Build

1. Detection Layer (30 Points): Process the CCTV video using any AI/Computer Vision models. Detect people, track movement, determine direction, and assign a unique session ID. Output structured JSON events.
2. Event Stream: Transmit structured events from the detection layer to the ingest API.
3. Intelligence API (35 Points): Build a REST API that ingests the events and exposes specific endpoints (/events/ingest, /metrics, /funnel, /heatmap, /anomalies, /health).
4. Live Dashboard (10 Bonus Points): Create a UI that updates a store metric in real-time.

4. Edge Cases You Must Handle

- Group Entry: Count 3 people walking in together as 3 individuals.
- Staff Exclusion: Identify staff and exclude them from customer metrics.
- Re-entry: Catch when a person steps outside and walks back in, logging it as a REENTRY rather than a new visit.
- Camera Overlap: Ensure a person seen on two cameras isn't double-counted.
- Queues & Occlusion: Track queue depth and handle partially hidden people.

5. Production Readiness (20 Points)

- Containerized: Must start with a single `docker compose up` command.
- Idempotent & Resilient: API shouldn't break if same data is sent twice, returns HTTP 503 instead of crashing on DB failure.
- Testing & Logging: >70% test coverage and structured logging.

6. AI Usage Policy (15 Points)

- Fully Open-Book: Expected to use AI (ChatGPT, Claude, Gemini, etc.).

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- Documentation: Include a DESIGN.md and CHOICES.md explaining architectural decisions and where AI helped (e.g., model selection, schema design). Include AI prompts in test files.

7. Submission & Evaluation

- Acceptance Gate: Graded only if docker compose up works, ingests events without crashing, and returns valid JSON.
- Follow-Up Interview: After submitting repo, you answer 5 contextual questions in a 30-min async video to prove understanding.